```
import pandas as pd
d_images.csv")
```

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0bd8c- 3-b5...	https://api.nga.gov/iiif/0000bd8c- 39de-4453-b5...
1668a- 8-92...	https://api.nga.gov/iiif/0001668a- dd1c-48e8-92...
2e210- 1-bb...	https://api.nga.gov/iiif/0002e210- 8bc2-4471-bb...
32658- 3-8b...	https://api.nga.gov/iiif/00032658- 8a7a-44e3-8b...

```
!pip install pandas pillow requests torch torchvision scikit-learn
```

```
Requirement already satisfied: setuptools in /usr/local/lib/python3
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Building wheels for collected packages: clip
  Building wheel for clip (setup.py) ... done
  Created wheel for clip: filename=clip-1.0-py3-none-any.whl size=
  Stored in directory: /tmp/pip-ephem-wheel-cache-_fnj_c97/wheels/
Successfully built clip
Installing collected packages: ftfy, ultralytics-thop, ultralytics,
```

```
import pandas as pd
import requests
import os
from PIL import Image
import torch
import clip
import numpy as np
import matplotlib.pyplot as plt
from sklearn.metrics.pairwise import cosine_similarity
from ultralytics import YOLO
import cv2
```

```
df = df[df["openaccess"] == 1.0]

print("Open access images:", len(df))
```

```
Open access images: 68394
```

```
df_sample = df.sample(200, random_state=42)

urls = df_sample["iiifthumburl"].tolist()
```

```
os.makedirs("images", exist_ok=True)

for i, url in enumerate(urls):

    try:
        img = requests.get(url).content
        with open(f"images/{i}.jpg", "wb") as f:
            f.write(img)
    except:
        print("Failed:", url)

print("Download complete")
```

```
Download complete
```

```
yolo_model = YOLO("yolov8n.pt")
```

```
Downloading https://github.com/ultralytics/assets/releases/download/
```

```
image_objects = {}
object_crops = {}
```

```
for file in os.listdir("images"):

    path = os.path.join("images", file)
    img = cv2.imread(path)

    results = yolo_model(path)

    labels = []
    crops = []

    for r in results:
        boxes = r.boxes.xyxy.cpu().numpy()
        classes = r.boxes.cls.cpu().numpy()

        for box, cls in zip(boxes, classes):

            label = yolo_model.names[int(cls)]
            labels.append(label)

            x1, y1, x2, y2 = map(int, box)
            crop = img[y1:y2, x1:x2]

            crops.append(crop)

    image_objects[file] = list(set(labels))
    object_crops[file] = crops
```

```
image 1/1 /content/images/68.jpg: 640x480 1 person, 6.0ms
Speed: 2.3ms preprocess, 6.0ms inference, 1.0ms postprocess per ima

image 1/1 /content/images/26.jpg: 512x640 (no detections), 7.2ms
Speed: 2.0ms preprocess, 7.2ms inference, 0.5ms postprocess per ima

image 1/1 /content/images/90.jpg: 640x544 3 persons, 6.7ms
Speed: 2.3ms preprocess, 6.7ms inference, 1.0ms postprocess per ima

image 1/1 /content/images/74.jpg: 640x640 1 person, 6.9ms
Speed: 2.7ms preprocess, 6.9ms inference, 1.0ms postprocess per ima

image 1/1 /content/images/121.jpg: 640x448 1 person, 6.0ms
Speed: 1.8ms preprocess, 6.0ms inference, 1.0ms postprocess per ima

image 1/1 /content/images/148.jpg: 640x416 (no detections), 6.0ms
Speed: 2.1ms preprocess, 6.0ms inference, 0.4ms postprocess per ima

image 1/1 /content/images/176.jpg: 640x512 (no detections), 6.1ms
Speed: 2.6ms preprocess, 6.1ms inference, 0.4ms postprocess per ima

image 1/1 /content/images/118.jpg: 640x448 1 person, 7.8ms
Speed: 2.3ms preprocess, 7.8ms inference, 1.1ms postprocess per ima
```

```
image 1/1 /content/images/56.jpg: 640x480 (no detections), 6.9ms
Speed: 2.2ms preprocess, 6.9ms inference, 0.5ms postprocess per ima

image 1/1 /content/images/16.jpg: 640x448 1 person, 5.9ms
Speed: 2.3ms preprocess, 5.9ms inference, 1.1ms postprocess per ima

image 1/1 /content/images/5.jpg: 640x416 (no detections), 6.1ms
Speed: 2.0ms preprocess, 6.1ms inference, 0.5ms postprocess per ima

image 1/1 /content/images/7.jpg: 640x448 1 keyboard, 6.3ms
Speed: 2.3ms preprocess, 6.3ms inference, 1.1ms postprocess per ima

image 1/1 /content/images/91.jpg: 640x544 1 potted plant, 7.0ms
Speed: 2.3ms preprocess, 7.0ms inference, 1.0ms postprocess per ima

image 1/1 /content/images/111.jpg: 480x640 3 persons, 6.1ms
Speed: 2.3ms preprocess, 6.1ms inference, 1.0ms postprocess per ima

image 1/1 /content/images/174.jpg: 480x640 (no detections), 5.5ms
Speed: 2.2ms preprocess, 5.5ms inference, 0.4ms postprocess per ima

image 1/1 /content/images/132.jpg: 640x480 (no detections), 6.2ms
Speed: 1.8ms preprocess, 6.2ms inference, 0.5ms postprocess per ima

image 1/1 /content/images/38.jpg: 640x448 (no detections), 6.2ms
Speed: 1.6ms preprocess, 6.2ms inference, 0.5ms postprocess per ima

image 1/1 /content/images/92.jpg: 640x416 1 person, 9.6ms
Speed: 2.7ms preprocess, 9.6ms inference, 1.3ms postprocess per ima

image 1/1 /content/images/171.jpg: 640x416 1 vase, 6.4ms
Speed: 2.1ms preprocess, 6.4ms inference, 1.4ms postprocess per ima

image 1/1 /content/images/83.jpg: 480x640 (no detections), 8.2ms
Speed: 2.6ms preprocess, 8.2ms inference, 0.5ms postprocess per ima
```

```
device = "cuda" if torch.cuda.is_available() else "cpu"

model, preprocess = clip.load("ViT-B/32", device=device)

print("CLIP model loaded")
```

```
100%|██████████████████████████████████████| 338M/338M [00:06<00:00
CLIP model loaded
```

```
object_embeddings = {}

for img_name, crops in object_crops.items():

    embeddings = []

    for crop in crops:

        crop = cv2.cvtColor(crop, cv2.COLOR_BGR2RGB)
        pil_img = Image.fromarray(crop)

        image = preprocess(pil_img).unsqueeze(0).to(device)

        with torch.no_grad():
            emb = model.encode_image(image)

        embeddings.append(emb.cpu().numpy())

    if embeddings:
        object_embeddings[img_name] = np.vstack(embeddings)
```

```
def compute_similarity(emb1, emb2):

    sim = cosine_similarity(emb1, emb2)

    return np.max(sim)
```

```
def find_similar(query_img, top_k=5):

    query_objects = set(image_objects[query_img])
    query_emb = object_embeddings[query_img]

    scores = []

    for img, emb in object_embeddings.items():

        if img == query_img:
            continue
        if len(query_objects.intersection(image_objects[img])) == 0
            continue

        score = compute_similarity(query_emb, emb)

        scores.append((img, score))

    scores.sort(key=lambda x: x[1], reverse=True)

    return scores[:top_k]
```

```
results = find_similar("10.jpg")

for img, score in results:
    print(img, score)
```

```
130.jpg 0.8568492993344421
53.jpg 0.8477820895838513
4.jpg 0.8256557798658666
82.jpg 0.7790539472680265
85.jpg 0.7680371881841006
```

```
def show_results(query):

    similar = find_similar(query)

    plt.figure(figsize=(12,3))

    query_img = cv2.imread(f"images/{query}")
    query_img = cv2.cvtColor(query_img, cv2.COLOR_BGR2RGB)

    plt.subplot(1,6,1)
    plt.imshow(query_img)
    plt.title("Query")
    plt.axis("off")

    for i,(img,score) in enumerate(similar):

        im = cv2.imread(f"images/{img}")
        im = cv2.cvtColor(im, cv2.COLOR_BGR2RGB)

        plt.subplot(1,6,i+2)
        plt.imshow(im)
        plt.title(f"{score:.2f}")
        plt.axis("off")

    plt.show()
```

```
show_results("10.jpg")
```

```
similarity_scores = []

for img, emb in object_embeddings.items():

    scores = []

    for img2, emb2 in object_embeddings.items():

        if img == img2:
            continue

        scores.append(compute_similarity(emb, emb2))

    similarity_scores.append((img, np.mean(scores)))
```

```
outliers = sorted(similarity_scores, key=lambda x: x[1])[:5]

print("Potential outliers:", outliers)
```

```
Potential outliers: [('131.jpg', np.float64(0.5059168806985979)), ('
```